



Question(s): 5/2

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E-mail: Rahul2.joshi@ril.com**Abstract:** This contribution describes how can the network operations vision be extended to include customer experience as a key guiding factor. Through big data analytics and instrumentation, individual customer experience records can be aggregated, analyzed and correlated with network events to generate automated, data-driven actions.**1. Introduction**

The operator community recognizes the importance of data-driven fully automated decision making in network operations, particularly in the light of limitations of traditional approaches to network management that often involve multiple systems and manual processes. Such limitations can result in delays, errors, and a reduction in customer satisfaction.

To address this challenge, Reliance Jio has undertaken the objective of integrating customer experience data into its network action decision-matrix. This approach enables us to evaluate the efficacy of every field action from a 'customer upwards view' and ensure that our network performance is aligned with the needs of our customers.

This initiative has resulted in an improved customer experience, and we remain committed to enhancing our network operations to meet evolving needs of our customers.

2. The Customer Close Looping (CCL) Initiative

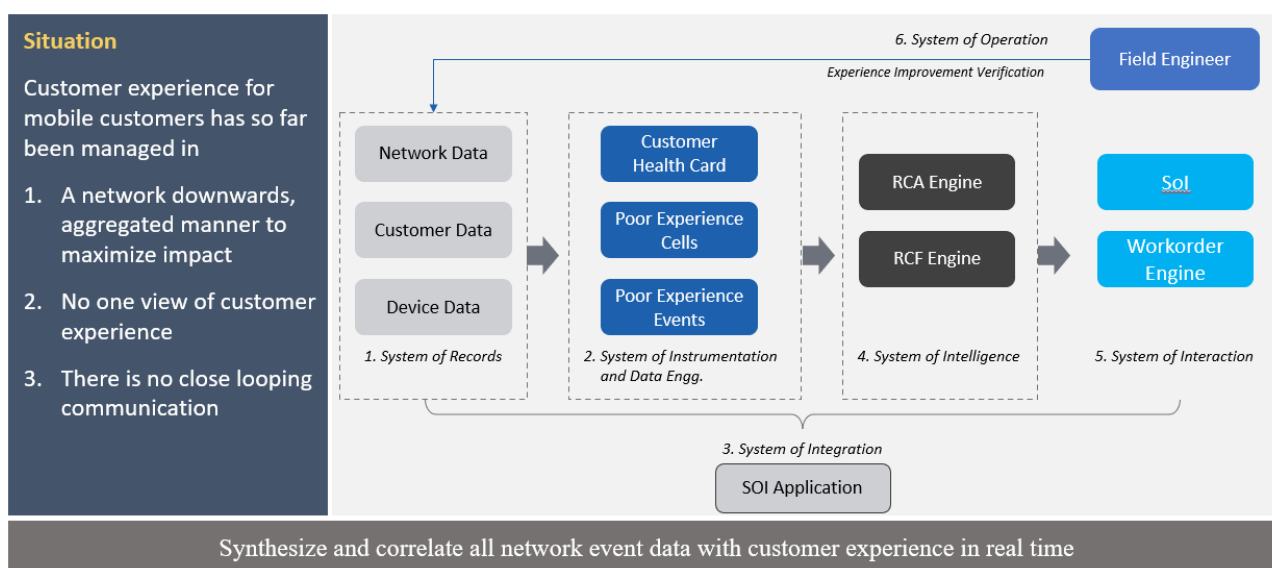
Reliance Jio launched the Customer Close Looping initiative encompassing multiple development tracks with cross-functional teams. The CCL initiative had following vision:

- Synthesize and correlate all network event data with customer experience in real time
- Empower field manpower with data analytics
- Using a and b, generate intelligent workflows that improve customer experience

The above vision was translated into multiple projects for end-to-end empowerment of operations manpower, some of which are listed below:

- a) Ingestion and correlation of various data streams into a common Big Data Lake
- b) Creation of algorithms to identify the influence of various network events in customer service experience journey
- c) Codification of algorithms, testing and automation of identification of customer experience degradation and its causes
- d) Derivation of simplified actions for customer experience improvement
- e) Generation and assignment of automated 'direct to doer' work orders

CCL Framework



All the above information is visualized in a simple format through a unified App called 'System of Interaction'. Further, each information is sliced according to the local geography area and when a field engineer logs to the App, he sees only the localized information relevant to him.

In this way, the System of Interaction (SoI) App became an all-in-one tool for field engineers. This unique ability to close the loop between network operations and customer experience, through a centralized app, sets it apart from other solutions in telecom industry.

3. Development tracks under Customer Close Looping Initiative:

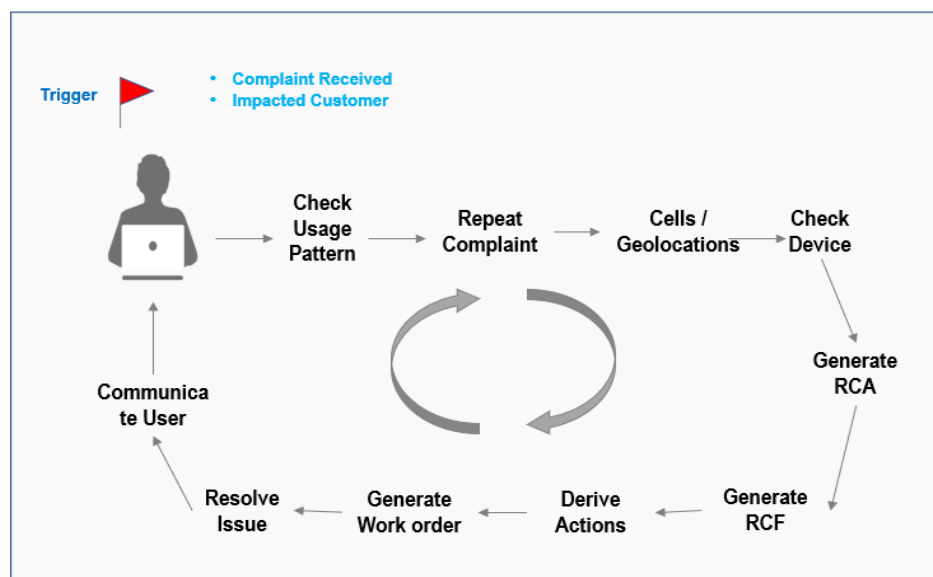
To enable a customer experience-centric view of network operations, a new approach was adopted for all actions related to network performance improvement, network optimization, and network build. First, various legacy data streams were ingested into a central Big Data Lake through data engineering efforts. Thereafter, multiple development tracks were launched, each resulting in its own innovative solution:

- a) The first track involved the development of the Customer Health Card, which correlates all events that can impact customer experience, such as network outage, network performance degradation, network capacity, network interference, etc in various environments. This

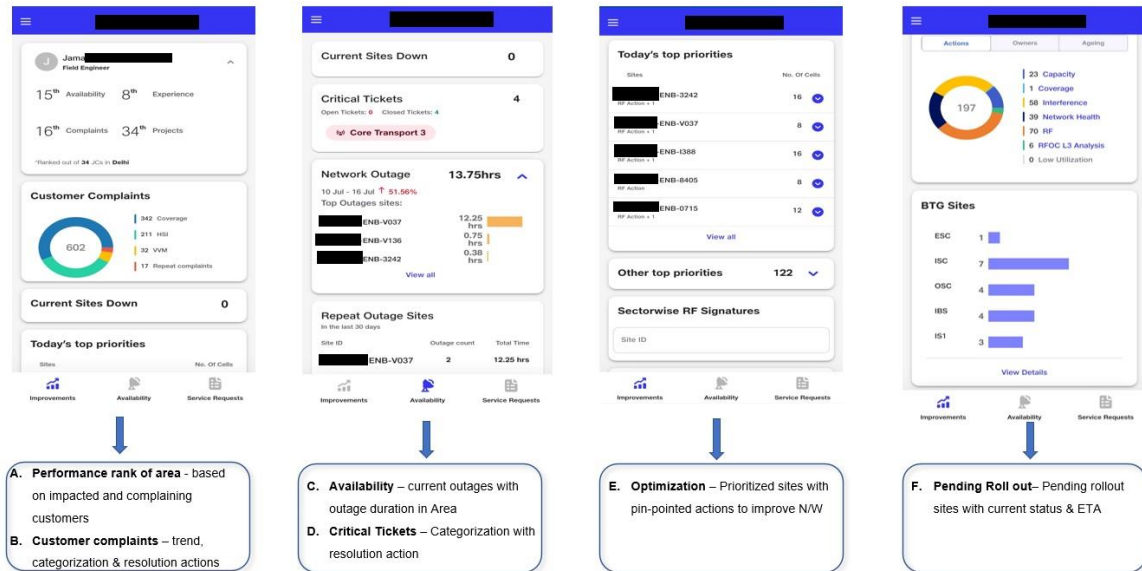
information is aggregated into a per-customer per day basis for the entire subscriber base, providing a clear indication of how each customer is experiencing the network. It also facilitates the aggregation of customer experience over a network topology.

- b) In the second track, the Root Cause Analysis (RCA) Engine was developed, that correlates customer experience with network events and network performance to identify the underlying causes of poor experience. It identifies the network nodes and radio sectors that caused the poor experience to customers.
- c) The third track involved development of the Root Cause Fix (RCF) Engine, that correlates the output of Root Cause Analysis (RCA) engine with prescribed fixes in the form of work order actions. These actions could be physical actions that are required to be done by the field workforce or digital actions that are done by the centralized NOC. The complete process of diagnosing events and generating work orders is automated through our Big Data Analytics capabilities.
- d) The fourth track was for development of the Alarm Categorization Engine, which synthesizes all faults/alarms/outages across the network topology for radio, fiber, and IP network assets and categorizes them for a cluster of customers in a geography. It also distributes each alarm into actionable categories, provides simplified information for diagnosis and troubleshooting, and suggests actions to resolve the alarm.
- e) The fifth track involved the creation of the Customer SR Correlation Engine, which analyses a customer complaint (service request) and deep dives into all network and customer events that caused it. It then extracts the root cause fix. The Customer SR Correlation Engine works over the Customer Health Card and RCF Engine.

Representative Flow



- f) Finally, the System of Interaction (SoI) was built that visualizes and makes the above four innovations accessible to the last-mile field team through a unified application on handheld device. It hides all complexity and presents simplified prioritized actions to the field engineer. The SoI empowers the field team to perform various analyses for network improvement, handle customer complaints, and assists in optimization and build workflows.



4. Proposal

- It is important to build a 'customer backwards' approach into network operations and network engineering. It is proposed that the network operations teams in the field get a customer experience-based view that enriches and empowers them to prioritize their network actions.
- Network actions should be seen and validated through customer experience. A system and method for binding customer experience management with network operations and engineering should be built
- Capability for automated and proactive network actions should be included in the network operations workflows.